

Exploring the Chameleon Effect in Metaverse Environments for Enabling Creative Communication

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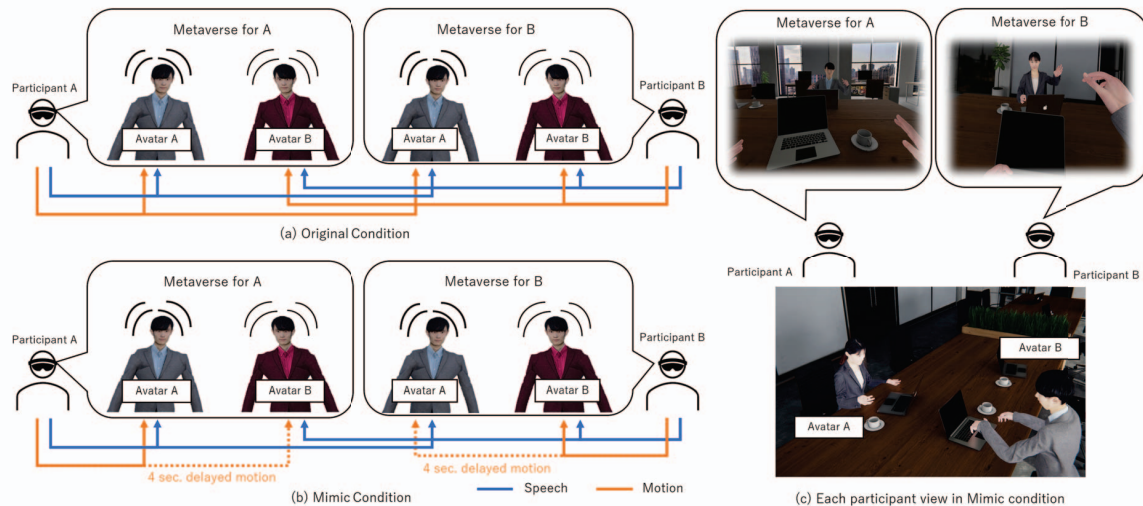


Figure 1: (a) In the original condition, each participant interacts with the other via an avatar that performs each motion as is. (b) In the mimic condition, each participant interacts with the other via the other avatar that mimics his or her own motion with a 4-second delay. (c) In the mimic condition, each participant interacts while observing the behavior of the other avatar with Chameleon Effect, even though they are actually performing different actions.

ABSTRACT

In this study, we explore the ‘Chameleon Effect’ within metaverse environments and its impact on creative communication. This experiment investigates how avatar mimicry in online interactions influences creativity. Participants were divided into Mimic and Original conditions, participating in brainstorming tasks, notably the creative tasks such as the Alternative Uses Task. The study assesses the quality of ideas generated—focusing on Fluency, Flexibility, Elaboration, and Originality—and participant impressions of their avatars. Our findings suggest significant implications for enhancing creativity in virtual settings, providing insights into the future of collaborative innovation in the metaverse.

Index Terms: Human-centered computing—Interaction design—Empirical studies in interaction design—

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1 INTRODUCTION

In recent years, the metaverse has attracted a great deal of attention in the business and entertainment fields, because it allows communication to overcome the constraints of social existence as well as physical existence. Metaverse interaction through avatars has the potential to improve the quality of communication, which was not possible with real face-to-face communication, because the avatar’s appearance, facial expressions, and motions can be changed dynamically. This study focuses on the avatar mimicry behavior in the metaverse, known as the ‘Chameleon Effect’ [1], and its impact on creativity. The Chameleon Effect, a psychological phenomenon of unconsciously imitating another’s behavior or attitude, is known to influence interpersonal relations and the quality of communication.

2 RELATEDWORKS

There are numerous benefits to achieving creative communication, which is vital in various contexts, such as business meetings and educational settings. Efforts to enhance creativity have been a long-standing focus of research. Previous research has shown that applying facilitation rules [5] and sharing personal stories before discussions [6] can enhance creativity in subsequent dialogues.

The evaluation of creativity is often conducted through brainstorming experiments. Brainstorming, developed by Alex Osborn [4], typically involves measuring the quantity and diversity of ideas generated during creative communication experiments.

However, challenges in interpersonal communication, such as production blocking, evaluation apprehension, and social loafing, have been identified as impediments to effective brainstorming [6].

This has led to increased attention on interactions within metaverse spaces.

In the metaverse, avatar-mediated communication allows for a range of expressions, potentially improving communication efficiency and fostering creativity. It is also known that the metaverse, with its limited non-verbal information and social cues, reduces the pressure from others.

In this context, the Chameleon Effect, or the mimicry of another's behavior, becomes significant. Research by Bailenson and Yee [1] demonstrated that avatars mimicking head movements can enhance intimacy. Similarly, a study by Won [7] showed a correlation between the degree of mimicry in movements during brainstorming and creative scores. Based on these studies, this research explores the potential of avatar movement intervention in the metaverse to stimulate brainstorming sessions.

Given these considerations, our study focuses on the relationship between the Chameleon Effect and creativity in a metaverse environment.

3 BRAINSTORMING EXPERIMENT

3.1 Data Collection

In our experiment, we implemented a metaverse system to investigate whether the Chameleon Effect, where one's movements are mimicked with a four-second delay in avatar-mediated communication, could impact creativity. Each pair of participants, wearing head-mounted displays, engaged in a discussion within this virtual space as shown in Fig. 1.

In our study, participants (16 males, 2 females) were paired and assigned to either the Mimic or Original condition. They engaged in a creative task, the Alternative Uses Task (AUT) [2], where they were asked to generate as many unique uses as possible for everyday objects. Each pair completed two tasks, each lasting three minutes. The dialogues during these tasks were recorded and transcribed for analysis.

3.2 Evaluation Axes and Scoring Method

In the AUT, creativity was evaluated in terms of Fluency, Flexibility, Elaboration, Originality [3].

1. Fluency: The total number of distinct ideas generated.
2. Flexibility: The variety of idea categories.
3. Elaboration: The level of detail in each idea.
4. Originality: The uniqueness of ideas, with rare ideas receiving higher scores.

Two independent evaluators assessed the transcribed data to determine scores for Fluency, Flexibility, Elaboration, and Originality.

In addition, impression of each other's avatars were assessed using 13 different items, such as likability and relatability. These scores were then averaged to provide an overall impression score.

4 RESULTS AND DISCUSSION

The results as shown in Fig. 2 highlight significant insights into the impact of the Chameleon Effect in our metaverse environment on creativity. Our analysis focused on two conditions: Mimic and Original. The evaluations were based on Fluency, Flexibility, Elaboration, Originality, and Impression scores.

1. Fluency, Flexibility, and Originality: The Original condition showed a tendency towards higher scores in Fluency, Flexibility, and Originality compared to the Mimic condition. This suggests that participants in the Original condition generated a greater number of ideas, which were more varied and unique.
2. Elaboration: Conversely, the Mimic condition outperformed the Original in terms of Elaboration. This indicates that participants engaging in mimicry provided more detailed elaborations of their ideas, potentially fostering a richer dialogue.

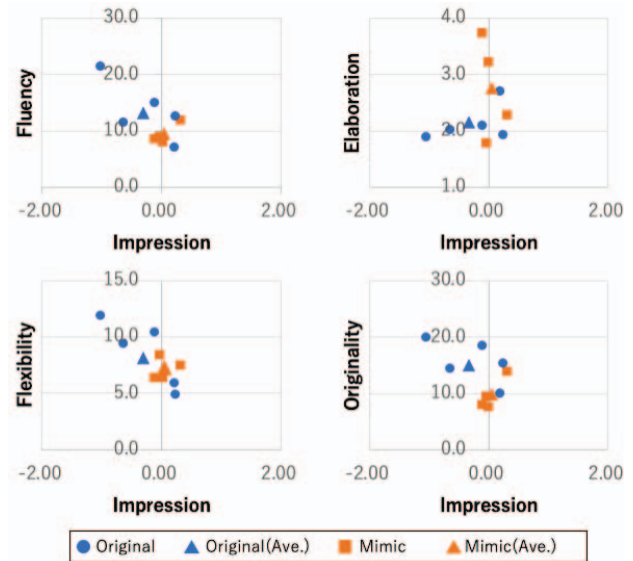


Figure 2: Relationship Between Each Score and Impression Score.

3. Impression Scores: Consistent with previous research [2], the Impression scores were higher in the Mimic condition than in the Original. This suggests that participants felt a more positive impression of their counterparts' avatars when mimicry was involved, possibly facilitating a more engaging and comfortable interaction.

The results indicate that the Original condition fosters a variety of creative ideas, while the Mimic condition deepens discussion, possibly leading to more refined ideas. This highlights the complex role of avatar mimicry in virtual creative collaboration.

5 CONCLUSION

This pilot study suggests potential influences of avatar mimicry on creative idea generation in the metaverse. Given the small sample size, these findings are preliminary and serve as a basis for more extensive future research. This work lays the groundwork for exploring effective communication strategies in virtual environments, leading towards enhanced creativity in the metaverse.

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